

## Linear inequalities



- 1
- |                 |                |                |                 |
|-----------------|----------------|----------------|-----------------|
| a A dashed line | b A solid line | c A solid line | d A dashed line |
| e A dashed line | f A solid line | g A solid line | h A dashed line |

- 2
- |              |                   |                    |                |
|--------------|-------------------|--------------------|----------------|
| a $x \geq 2$ | b $y \leq 5$      | c $y \geq -2x - 6$ | d $y < 2x + 6$ |
| e $y > -3x$  | f $y \leq 2x + 5$ |                    |                |

3 No

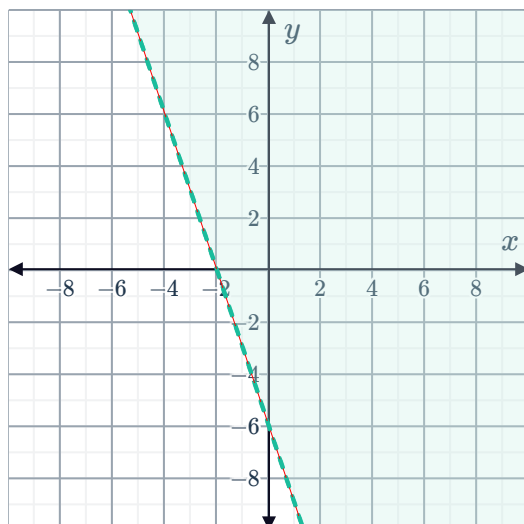
4 Yes

- 5
- |     |     |     |
|-----|-----|-----|
| a B | b C | c D |
|-----|-----|-----|

## Graphing linear inequalities

- 6
1. Replace the inequality symbol with an equals sign and graph the linear equation.
  2. Choose a test point from one of the half planes that is not on the line. Substitute its coordinates into the inequality.
  3. If a true statement results, shade the half plane containing the test point, if false then shade the other half plane.

- 7
- |                    |      |
|--------------------|------|
| a $x = -2, y = -6$ | b C  |
| c                  | d No |



8

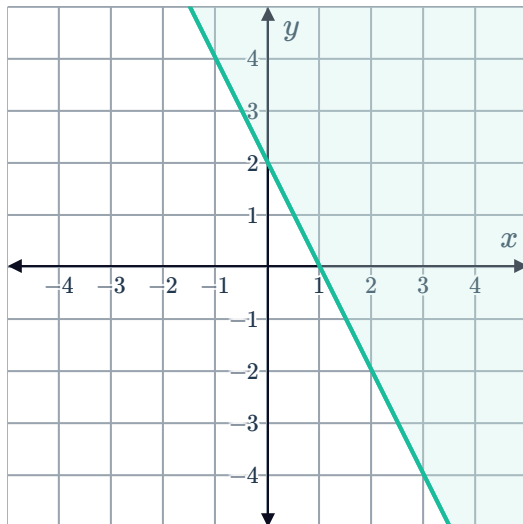
a  $x = 1, y = 2$



b B

c

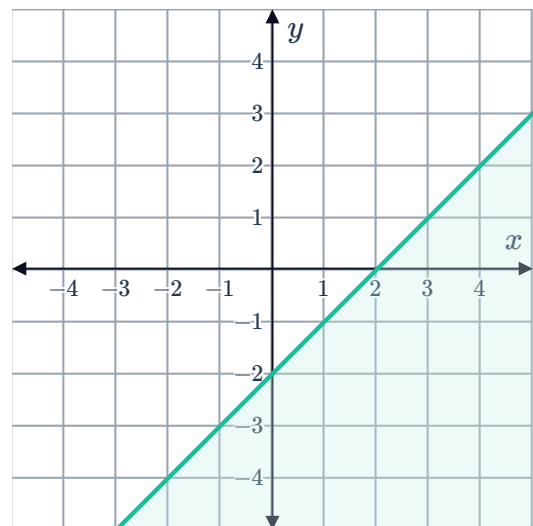
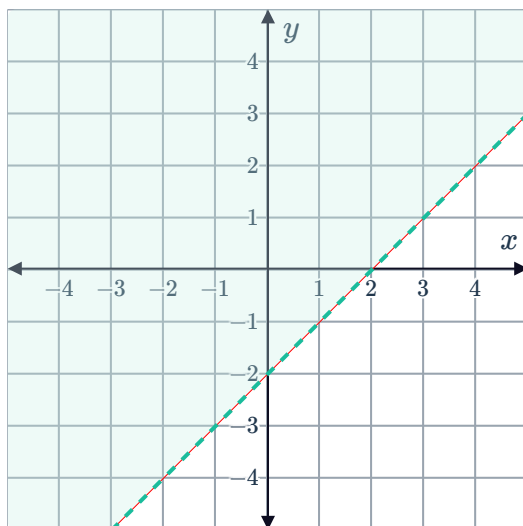
d Yes



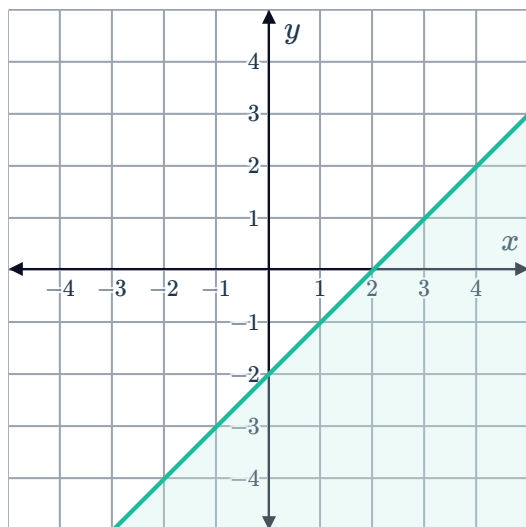
9

a

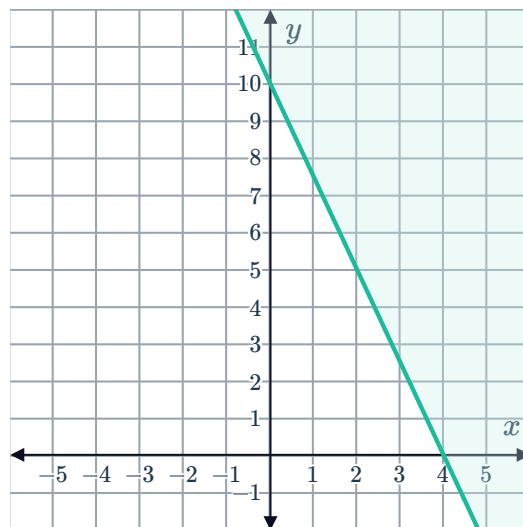
b



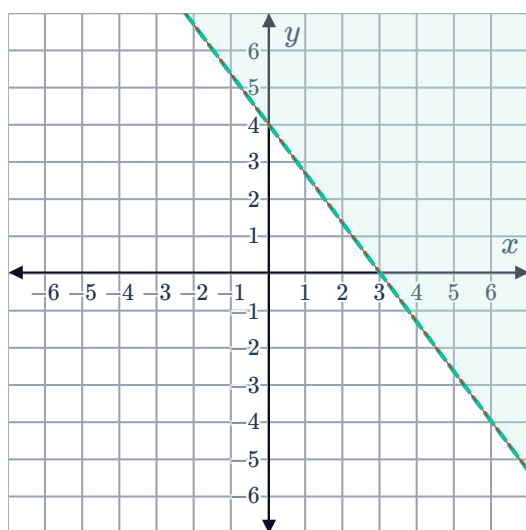
a



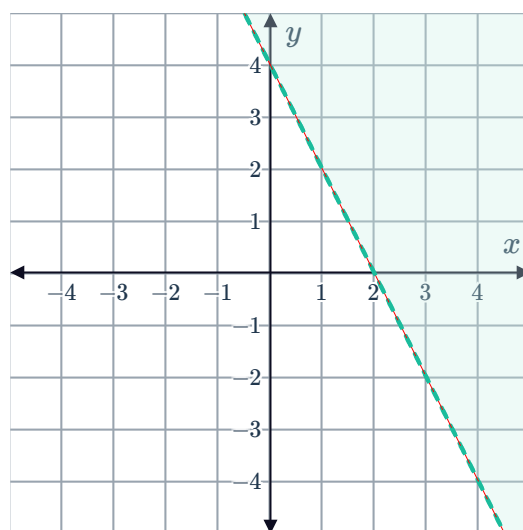
b



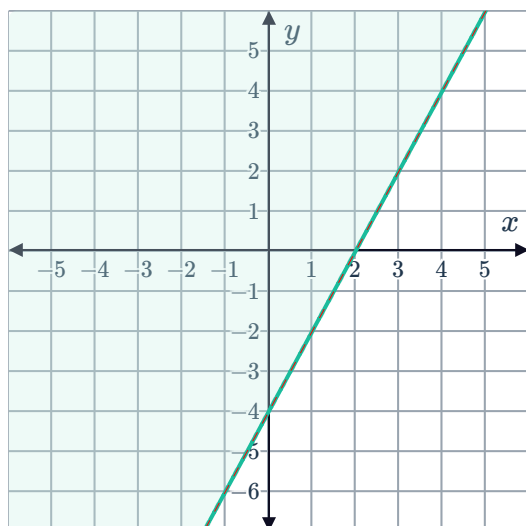
c



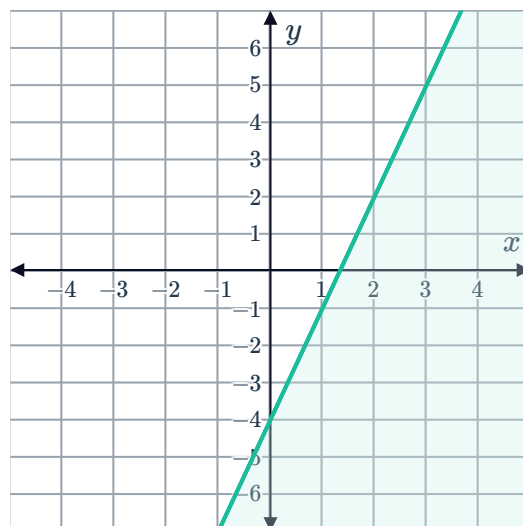
d



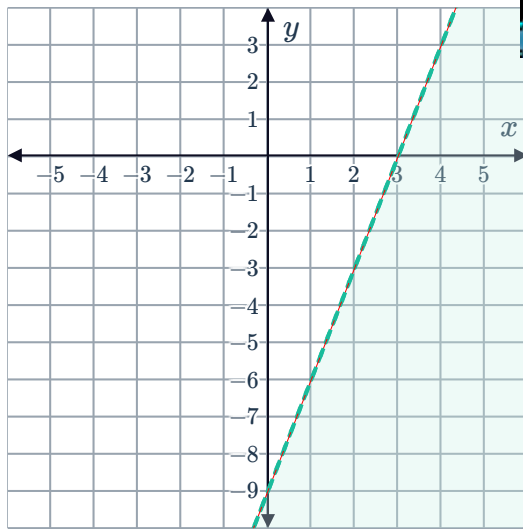
e



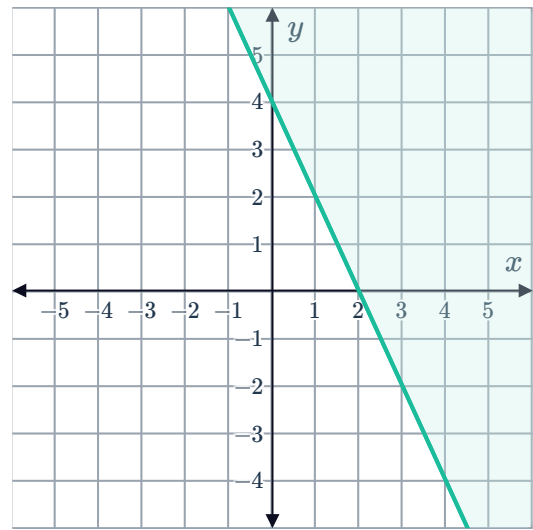
f



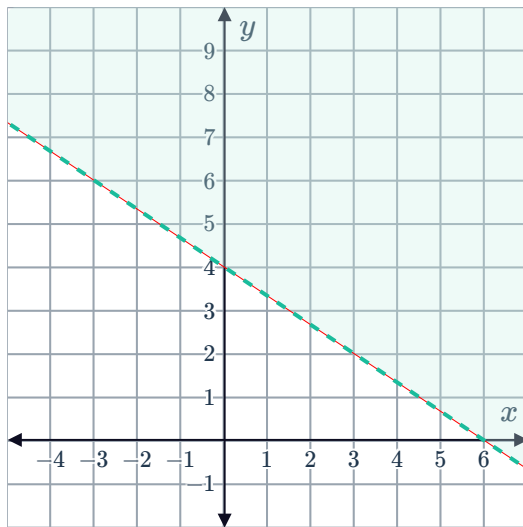
g



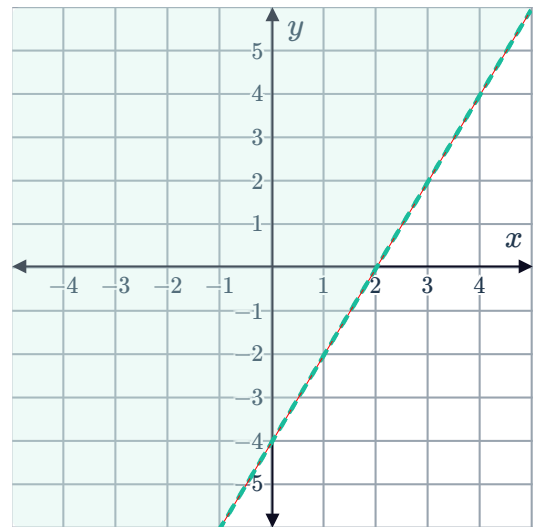
h



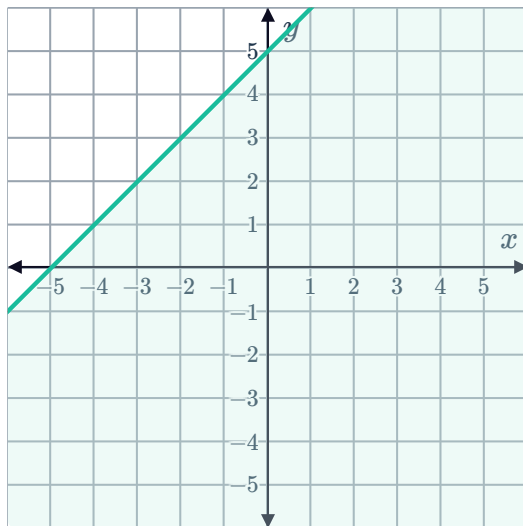
i



j



k



11

a  $y \leq 4x - 5$

b  $y \geq 3x - 6$

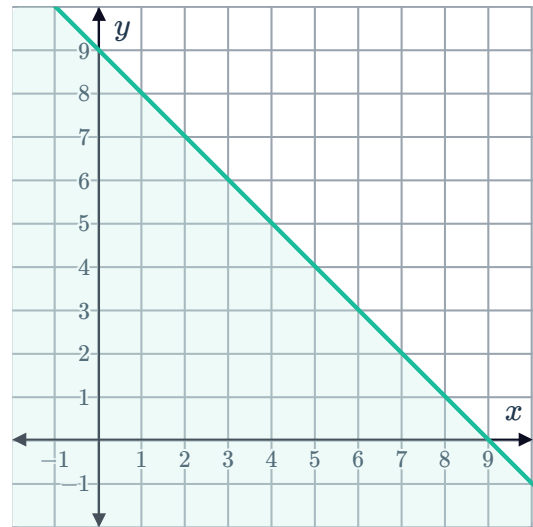


## Applications

12

a  $y \leq 9 - x$

b



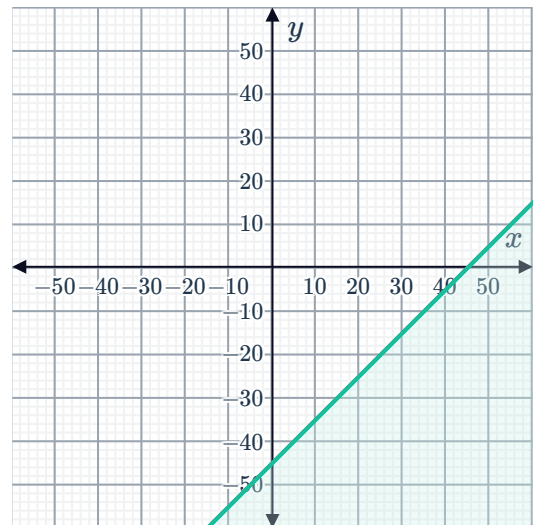
c

- i No    ii No    iii Yes    iv Yes

13

a  $y \leq x - 45$

b



c 65 minutes

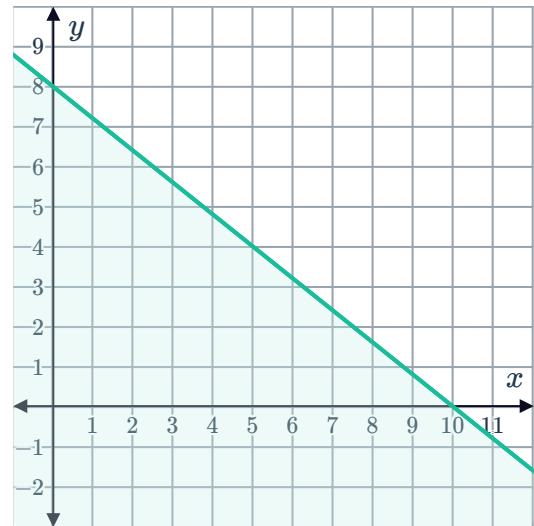
14

a

$$y \leq 8 - \frac{4}{5}x$$



b



c

- i No    ii Yes    iii No    iv Yes